

Signal Processing and Machine Learning (SPML) — Summer 2025

The new version of the course moves from an introduction to everything about ML to Generative Modelling or Generative AI in SP.

Tentative Outline:

Part I — Estimation and Regression

What is estimation vs. regression?
Similarities and differences, basics of estimation theory; basics of regression analysis; linear regression; nonparametric regression; neural network-based regression.

Part II — Generative Modeling

Generative Adversarial Networks;
Normalizing Flows;
Gaussian Mixture Models;

Variational Autoencoders; Diffusion Models.

Part III — Applications

Application in Parameter Estimation, e.g., in Wireless Communications.

Part I — Estimation and Regression

What is the difference between estimation and regression?

In both estimation and regression, we want to find out the relationship between variables.

Whereas estimation is only based on observations of random variables, for regression we need observations of pairs of **dependent** and **independent** random variables.

1. Estimation



↑
Statistical models by means
of explicit models $f_{X,Y}$ or
represented by data samples
 $(X_i, Y_i) \in S$, $|S| = s$.

— observed variable $X \in \mathbb{R}^n$, with

$$x_i \sim f_X(x_i) = \int f_{X,Y}(x_i, y) dy$$

if a statistical model exists.

— estimated variable $Y \in \mathbb{R}^m$,
with $y_i \sim f_{Y|X}(y_i|x)$

if a statistical model exists.

In case that a statistical model is partially or fully available, statistical signal processing (SSP) theory provides optimal solutions for the estimation problem.

Maximum Likelihood (ML) Estimation

In case that only partial knowledge $f_{X/Y}(x/y)$ is available, we obtain

$$\hat{y}_{ML} = \underset{y \in Y}{\operatorname{argmax}} \sum_{x_i \in Q} \log f_{X/Y}(x_i/y)$$

assuming the Q observations of the **unknown true** y ,

$$Q = \{x_1, x_2, \dots, x_Q\},$$

are i.i.d. ($\rightarrow$ refer to SSP course),

and further based on the assumption
 $x_i \sim f_{X|Y}(x_i|y)$.

Under mild conditions, ML estimators are **asymptotically efficient**, i.e., for $S \rightarrow \infty$, the estimator converges to the minimum achievable variance of the estimator ($\rightarrow$ Cramér Rao-bound).

Conditional Mean Estimation

In case of full knowledge of the statistical model, we obtain

$$\hat{y}_{CME} = \int y f_{Y|X_1 \dots X_Q}(y|x_1, x_2, \dots, x_Q) dy$$

$$\text{i.e. } y_{CME} = E_{Y|X_1 \dots X_Q} [Y | X_1 = x_1, \dots, X_Q = x_Q]$$

(→ conditional mean estimator)

In case that $Q=1$,

$$\begin{aligned}\hat{Y}_{CME} &= E_{Y|X} [Y|X] \\ &= \int_Y y f_{Y|X}(y|x) dy\end{aligned}$$

with $f_{Y|X}(y|x) = \frac{f_{X|Y}(x|y) f_Y(y)}{\int_Y f_{X|Y}(x|y) f_Y(y) dy}$

$\propto f_{X|Y}(x|y) f_Y(y)$
("proportional to")

and the given prior statistical model $f_Y(y)$ of the variable to be estimated.

The CME $\hat{Y}_{CME}$ is optimal in the sense

that it achieves the minimal mean squared error (MMSE) of the parameter to be estimated, i.e.,

$$\hat{y}_{\text{CME}} = \arg \min_{\hat{y} \in Y} E_{Y|X}[(\hat{y} - y)^2 | x]$$

$$\text{with } E_{Y|X}[(\hat{y} - y)^2 | x]$$

$$= \int_Y (\hat{y} - y)^2 f_{Y|X}(y|x) dy$$

($\rightarrow$ refer to the SSP course)

Example and Special Case

X and Y are jointly Gaussian distributed, i.e.,

$$Z = \begin{bmatrix} X \\ Y \end{bmatrix} \in \mathbb{R}^{n+m} \text{ with}$$

$$\mu_Z = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix} = \begin{bmatrix} E_x[X] \\ E_y[Y] \end{bmatrix} \text{ and}$$

$$C_Z = \begin{bmatrix} C_x & C_{xy} \\ C_{yx} & C_y \end{bmatrix} \quad \text{with}$$

$$C_x = \text{Var}[X] = E_x[(X - \mu_x)(X - \mu_x)^T]$$

$$C_y = \text{Var}[Y] = E_y[(Y - \mu_y)(Y - \mu_y)^T]$$

$$C_{xy} = \text{Cov}[X, Y] = E_{x,y}[(X - \mu_x)(Y - \mu_y)^T]$$

$$C_{yx} = C_{xy}^T, \text{ then}$$

$$\hat{y}_{\text{LMMSE}} = C_{yx} C_x^{-1} (x - \mu_x) + \mu_y$$

is a linear estimator, aka a linear minimum MSE estimator (LMMSE), which achieves the minimum MSE

$$\text{trace}[C_y - C_{yx} C_x^{-1} C_{xy}].$$

($\rightarrow$ refer to SSP course)

Conditional Prior

In many application and because of methodological purposes it can be assumed that the prior $f_Y(y)$ is available as

$$f_Y(y) = \int f_{Y|X}(y|x) f_X(x) dx$$

with a given $f_X(x)$ and a **conditional prior** $f_{Y|X}(y|x)$, which might be in a favorable shape. Consequently,

$$E_{Y|X}[Y|X] = E_{X|X}[E_{Y|X,X}[Y|X, x]|X]$$

Proof:

$$E_{X|X}[E_{Y|X,X}[Y|X, x]|X]$$

$$= \int_{\Delta} E_{Y/X, \delta} [Y/X, \delta] f_{\delta/X}(\delta/x) d\delta$$

$$= \int_{\Delta} \int_Y y f_{Y/X, \delta}(y/x, \delta) dy f_{\delta/X}(\delta/x) d\delta$$

$$= \int_Y y \int_{\Delta} f_{Y, \delta/X}(y, \delta/x) d\delta dy$$

$$= \int_Y y f_{Y/X}(y/x) dy$$

$$= E_{Y/X}[Y/X].$$

Given only a conditional prior, the CNE is obtained as

$$y_{CNE} = E_{\delta/X} \left[\int_Y y f_{Y/X, \delta}(y/x, \delta) dy \middle| x \right]$$

In case that the conditional prior

is Gaussian, or can be selected as a model, the CME estimator appears to be the expected mean LMMSE estimator

$$\hat{y}_{\text{CME}} = E_{\delta|x} [\hat{y}_{\text{LMMSE}}(x, \delta) | x],$$

i.e. it's an average of linear estimators.

